



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

29-04-2023

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	C	B	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,D	A,C	A,B,C,D	A,C	A,B,C	A,C	A,B,D	B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	4.00	20.00	2.80	2.50	0.14 to 0.16	1.00		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	D	C	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,C	B,C	A,C,D	A,B,C,D	A,B	A,B,C,D	A,B,C,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	4.04	1.00	7.00	2.00	10.00	7.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	B	C	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,C,D	A,C,D	A,C,D	A,C	A,D	B	A,B,D	B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	7.00	11.00	11.00	4.00	10.00	1.00		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)

If resistance of rheostat is increased current will decrease. Now since current decreases, so will the potential difference across each branch and hence voltmeter reading will increase.

2. Ans (C)

$$f_{x_{\text{net}}} = mg \cos 30^\circ$$

$$\therefore a_x = g \cos 30^\circ$$

3. Ans (B)

For $0 \leq t \leq \frac{1}{2}$, body will perform SHM with time period 2 sec starting from -ve amplitude.

$\therefore$ At $t = \frac{1}{2}$ sec, body passes through $x = 0$ with speed 2π m/sec.

$\therefore$ At $x = \frac{\pi}{2}$, we have

$$v^2 = 4\pi^2 + 2 \times 5\pi \times \frac{\pi}{2} \quad (v^2 = u^2 + 2as)$$

$$\Rightarrow v = 3\pi \text{ m/sec}$$

4. Ans (B)

$$\phi = -\frac{1}{2}R^2\theta B_1 + \frac{1}{2}B_2 3R^2\theta$$

$$\therefore \frac{d\phi}{dt} = \epsilon = \frac{R^2\theta}{2} \frac{dB_1}{dt} - \frac{3R^2\theta}{2} \frac{dB_2}{dt}$$

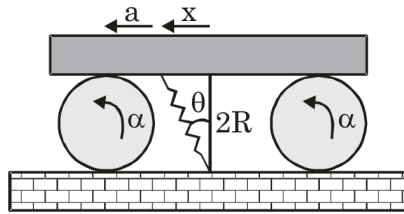
$$= \frac{R^2\theta}{2} \cdot 2K_0 - \frac{3R^2\theta}{2} \cdot 4K_0$$

$$\therefore \text{emf} = 5T^2\theta K$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,B,D)



Tension in the spring = kR

Horizontal Force on the plank

$$= -kR \sin \theta$$

$$= -kR \times \frac{x}{2R} = -\frac{kx}{2}$$

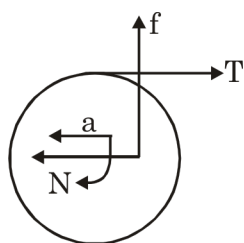
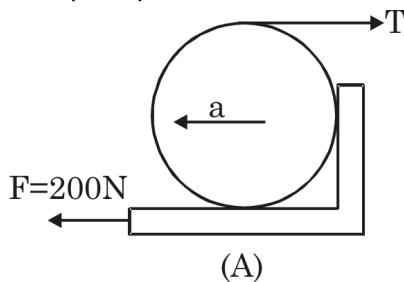
$$a = -\frac{kx/2}{M + 2 \times (M + I_{CM}/R^2) \times 1/4}$$

$$a = -\frac{kx/2}{M + 2 \times (M + M/2) \times 1/4}$$

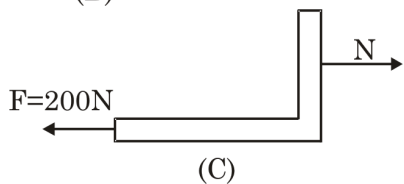
$$a = -\frac{2kx}{7M} = -\omega^2 x$$

$$T = 2\pi \sqrt{\frac{7M}{2k}} = 0.2s$$

6. Ans (A,C)



(B)



(C)

For L shaped massless body

$$F - N = 0 \Rightarrow N = 200N \quad \text{----- (A)}$$

For the system

$$200 - T = 4a \quad \text{----- (B)}$$

For cylinder

$$TR - fR = I\alpha$$

$$T - \mu N = I\alpha/R^2$$

$$T - 0.2 \times 200 = Ma/2 = 2a$$

$$T - 40 = 2a \quad \text{----- (C)}$$

From (B) & (C)

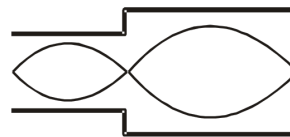
$$160 = 6a$$

$$a = 80/3 \text{ m/s}^2$$

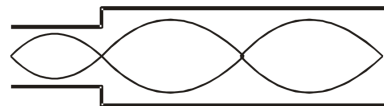
$$T = 40 + 160/3 = 280/3 \text{ N}$$

7. Ans (A,B,C,D)

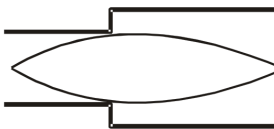
Velocity of sound in 2nd tube is twice of the 1st tube thus the wave length. Possible configuration may be:



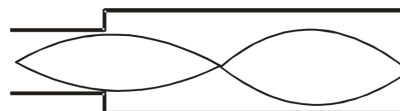
Join is pressure node $y/x=2$



Join is pressure node $y/x=4$



Join is pressure node $y/x=2$



Join is pressure node $y/x=6$

8. Ans (A,C)

$$y = \frac{n\lambda D}{d}$$

$$v_f = \frac{dy}{dt} = \frac{n\lambda}{d} \frac{dD}{dt} = \frac{n\lambda D}{dD} \frac{dD}{dt} = \frac{y}{D} v_{\text{mean}}$$

$$= \frac{5}{50} \times (100 \times 1) = 10 \text{ cm/s}$$

9. Ans (A,B,C)

Initially :

$$P_1 A L_1 = n R T_1$$

$$P_1 A = \frac{n R T_1}{L_1}$$

$$\frac{\theta^2}{2 A \epsilon_0} = \frac{n R T_1}{L_1}$$

Finally :

$$P_2 A L_2 = n R T_2$$

$$P_2 A = \frac{n R T_2}{L_2}$$

$$\frac{\beta^2 Q^2}{2 A \epsilon_0} = \frac{n R T_2}{L_2}$$

By energy conservation :

$$\begin{aligned} \frac{1}{2} \epsilon_0 E^2 A L_1 + \frac{f}{2} n R T_1 \\ = \frac{1}{2} \epsilon_0 E_2^2 A L_2 + \frac{f}{2} n R T_2 \end{aligned}$$

10. Ans (A,C)

$$y = a x^2; \quad y = \frac{H}{R^2} x^2$$

$$m g - \rho_L \frac{1}{2} \pi x^2 y g = m \frac{d v}{d t}$$

$$V \cdot \frac{d v}{d y} = \frac{d v}{d t} = g - \frac{\rho_L}{\rho_s} \frac{x^2}{R^2} \times \frac{y g}{H}$$

$$\int V d v = \int \left(g - \frac{\rho_L}{\rho_s} g y^2 \right) d y$$

11. Ans (A,B,D)

Let charge at (-1) be q_1 & at (+1) be q_2 .

$$\therefore \frac{K q_1}{0.5} + \frac{K q_2}{1.5} = \frac{K q_1}{1} + \frac{K q_2}{1} [V = 2.7 \text{ volt}]$$

$$\Rightarrow 2 q_1 + \frac{2}{3} q_2 = q_1 + q_2$$

$$\Rightarrow q_1 = \frac{q_2}{3} \Rightarrow \frac{q_1}{q_2} = \frac{1}{3}$$

Both the charges are positive.

12. Ans (B,C)

We have,

$$e V = \frac{h c}{\lambda} - \phi$$

$$\Rightarrow \frac{1}{\lambda} = \frac{e V}{h c} + \frac{\phi}{h c} \quad [y = m x + c]$$

PART-1 : PHYSICS

SECTION-II

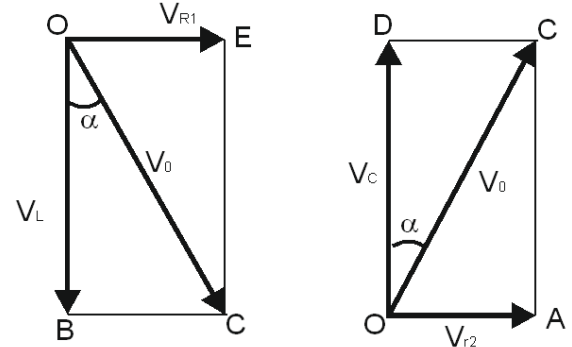
1. Ans (4.00)

$$\omega_b = \Delta V_E + \Delta U_{\text{surface}}$$

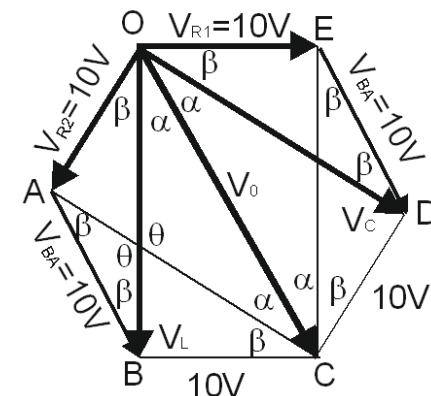
$$\frac{\epsilon_r \epsilon_0 A V^2}{d} = \frac{1}{2} \frac{\epsilon_r \epsilon_0 A V^2}{d} + \sigma \cdot 2 A$$

$$V = \sqrt{\frac{4 \sigma d}{\epsilon_0 \epsilon_r}}$$

2. Ans (20.00)



Overlapping both phasor diagrams



In trapezium ABCO

$$2\alpha + \theta = 2\beta + \theta$$

$$\alpha = \beta$$

In rectangle OACD

$$2\alpha + \beta = 90^\circ$$

$$3\alpha = 90^\circ$$

$$\alpha = 30^\circ$$

Thus hexagon OABCDE is regular and

$$V_0 = 10 \times 2 = 20 \text{ V}$$

3. **Ans (2.80)**

The maximum possible acceleration of the top block without slipping is $m_b a_{\max} = \mu m_t g$. If the top block is not slipping then the angular frequency is given by

$$\omega_2 = \sqrt{\frac{k}{m_t + m_b}},$$

so $a_{\max} \geq A\omega_2^2$

$$\text{or } A_c = \mu g \frac{m_t}{k} \left(1 + \frac{m_t}{m_b} \right)$$

$$= 0.5 \times 10 \times \frac{2}{500} \left(1 + \frac{2}{5} \right)$$

$$= \frac{7}{250} \text{ m}$$

$$= 2.8 \text{ cm}$$

4. **Ans (2.50)**

Work performed in one cycle = area under the curve

From the given information,

$$W = \frac{P_0 V_0}{4} = \frac{R \lambda 300}{4}$$

$$\therefore \text{Heat rejected/cycle} = \frac{100R}{4} = 25R$$

$$\therefore \Delta T/\text{Cycle} = \frac{25R}{1 \times 10R} = 2.5$$

5. **Ans (0.14 to 0.16)**

Let m_1 = mass of the iron rod, m_2 = mass of mercury, l_0 = length of the iron rod at 0°C , L_0 = height of mercury at 0°C , A_0 area of cross-section of the glass tube at 0°C . The distance of the centre of mass from the point of suspension at 0°C is given by

$$x_{\text{cm at } 0^\circ\text{C}} = \frac{m_1 \frac{l_0}{2} + m_2 \left(l_0 - \frac{L_0}{2} \right)}{m_1 + m_2}$$

$$\left(\because x_{\text{cm}} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} \right)$$

$$\text{Similarly } x_{\text{cm at } t^\circ\text{C}} = \frac{m_1 \frac{L_t}{2} + m_2 \left(l_t - \frac{L_t}{2} \right)}{m_1 + m_2}$$

In order that the centre of mass of the pendulum may not rise or fall.

$$x_{\text{cm at } 0^\circ\text{C}} = x_{\text{cm at } t^\circ\text{C}}$$

$$\text{or } m_1 \frac{l_0}{2} + m_2 \left(l_0 - \frac{L_0}{2} \right) = m_1 \frac{L_t}{2} + m_2 \left(l_t - \frac{L_t}{2} \right)$$

When the rod is light $m_1 = 0$

$$\therefore m_2 \left(l_0 - \frac{L_0}{2} \right) = m_2 \left(l_t - \frac{L_t}{2} \right) \text{ or } l_t - l_0 = \frac{1}{2}(L_t - L_0)$$

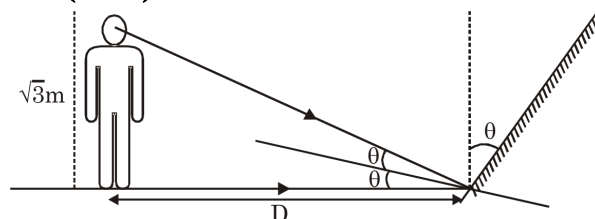
$$\text{or } l_0 \alpha_i t = \frac{1}{2} L_0 (\gamma_m - 2\alpha_g) t$$

where suffix i is for iron, m for mercury and g for glass.

$$\therefore L_0 = 2l_0 \frac{\alpha_i}{\gamma_m - 2\alpha_g}$$

$$\Rightarrow L_0 = \frac{2 \times 1 \times 12 \times 10^{-6}}{182 \times 10^{-6} - 2 \times 9 \times 10^{-6}} = 0.146 \text{ m} \approx 0.15 \text{ m}$$

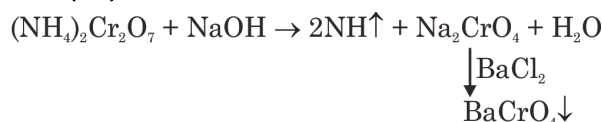
6. **Ans (1.00)**



PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (C)**



2. **Ans (D)**

XF_4 type molecule have zero dipole moment when it is either tetrahedral or square planar.

4. **Ans (D)**

(A) Victor Mayer's Test is given by Halide and Alcohol both

(B) Beilstein test is given by Halogen containing compound

(C) Monomer of polyethene is ethene

(D) It is a test to distinguish fructose and glucose

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (A,B,C)**

Since cis form is polar

$\therefore$ more soluble in polar solvent while trans form is more soluble in non polar solvent

6. **Ans (B,C)**

It is neutral.

NO gas is not isoelectronic with superoxide

Exhibit resonance.

7. **Ans (A,C,D)**

It looks as if process is reversible as it is carried out in small steps.

But $P_{\text{ex}} = 0$, So $w = 0$.

The wall is diathermic, so process is isothermal.

Hence, $T_1 = T_2$.

So, $\Delta U = 0$

So, $U_2 = U_1$

As gas has expanded without change in temperature, so pressure has decreased.

So, $q = 0$

8. **Ans (A,B,C,D)**

Let P_A^* and P_B^* be the vapor pressures of pure liquids and let n_A and n_B be the moles of A and B in starting liquid.

Hence

$$900 = P_A^* \times \frac{n_A}{n_A + n_B} + P_B^* \times \frac{n_B}{n_A + n_B} \quad \dots (1)$$

$$\text{Total moles of residue} = \frac{1}{3}(n_A + n_B);$$

$$n_A(\text{residue}) = 0.3 \times \frac{1}{3}(n_A + n_B) = \frac{1}{10}(n_A + n_B)$$

$$\text{Total moles in condensate} = \frac{2}{3}(n_A + n_B)$$

$$\Rightarrow n_A(\text{condensate}) = 0.6 \times \frac{2}{3}(n_A + n_B) = \frac{2}{5}(n_A + n_B)$$

$$\Rightarrow n_A(\text{total}) = \frac{1}{10}(n_A + n_B) + \frac{2}{5}(n_A + n_B)$$

$$\Rightarrow \frac{n_A}{n_A + n_B} = \frac{1}{2}$$

$$\Rightarrow \text{Initial } x_A = 0.5; \quad x_B = 0.5$$

$$\Rightarrow 900 = \frac{P_A^*}{2} + \frac{P_B^*}{2} \quad \dots (2)$$

$$P_T \text{ over residue} = 860$$

$$\Rightarrow 860 = P_A^* \times 0.3 + P_B^* \times 0.7 \quad \dots (3)$$

Solving (2) and (3) gives

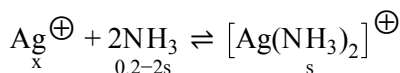
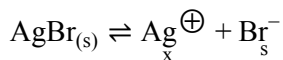
$$P_B^* = 800 \text{ torr}$$

$$P_A^* = 1000 \text{ torr}$$

$$P_T \text{ over condensate} = 1000 \times 0.5 + 800 \times 0.4$$

$$= 600 + 320 = 920 \text{ torr}$$

9. **Ans (A,B)**



$$x \times s = 2 \times 10^{-10}$$

$$x \times s \frac{s}{x(0.2 - 2s)^2} = 16 \times 10^{-4}$$

$$\frac{s}{0.2 - 2s} = 4 \times 10^{-2}$$

$$\Rightarrow s = 8 \times 10^{-3} - 8 \times 10^{-2}s$$

$$\Rightarrow s = 8 \times 10^{-3}$$

$$x = \frac{10^{-7}}{4}$$

$$\text{Now, } \lambda_1 = \frac{K \times 1}{1000 \times x}$$

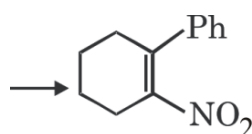
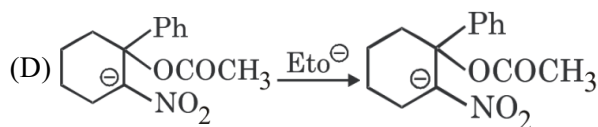
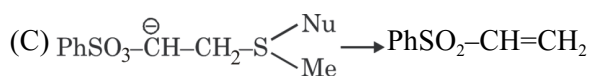
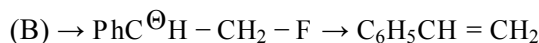
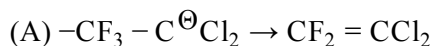
$$K_{\text{Ag}^+} = \lambda \times 1000 \times x$$

$$= 6 \times \frac{10^{-7}}{4}$$

$$= 1.5 \times 10^{-7} \text{ sm}^{-1}$$

$$= 64 \times 10^{-3} \text{ sm}^{-1}$$

10. **Ans (A,B,C,D)**



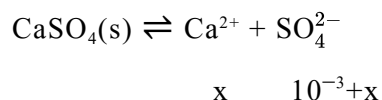
PART-2 : CHEMISTRY

SECTION-II

1. Ans (4.04)

$$\Lambda_{\text{Na}_2\text{SO}_4} = \frac{2.6 \times 10^{-4}}{10^{-3}} \times 1000 = 260$$

$$\Lambda_{\text{SO}_4^{2-}} = 260 - 2(50) = 160$$



$$7 \times 10^{-4} = 10^{-3} [2 \times 10^{-3} \times 50 + x \times 120 + (10^{-3} + x) \times 160]$$

$$\therefore x = \frac{11}{7} \times 10^{-3}$$

$$\text{Hence, } K_{\text{sp}} \text{ of } \text{CaSO}_4 = x \times (10^{-3} + x)$$

$$= 4.04 \times 10^{-6} \text{ M}^2$$

2. Ans (1.00)

Atoms on one face of unit cell = 2.

Total number of atoms one face of crystal

$$= 2 \times \frac{1}{6} \times 6 \times 10^{30} = 2 \times 10^{30}$$

$$\therefore \text{Unit cells at one face } N = \frac{1}{\sqrt{2}} \times 10^{15}; 10^{30} = N^2$$

$$\therefore \sqrt{2}a = 4(50)$$

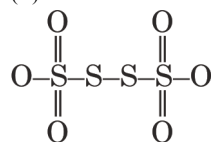
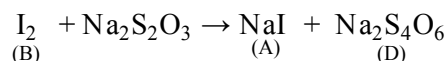
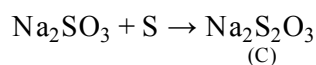
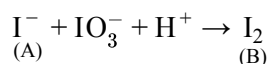
$$a = \frac{200}{\sqrt{2}} \text{ nm}$$

Edge length of crystal

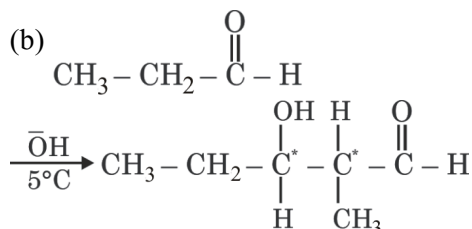
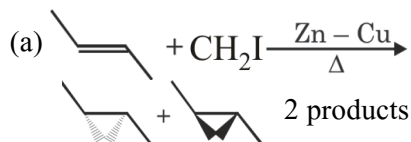
$$= \left(\frac{200}{\sqrt{2}} \times 10^{-9} \text{ m} \right) \times \frac{1}{\sqrt{2}} \times 10^{15} = 10^8 \text{ m}$$

$$\therefore \text{Area} = a^2 = 10^{16} \text{ m}^2$$

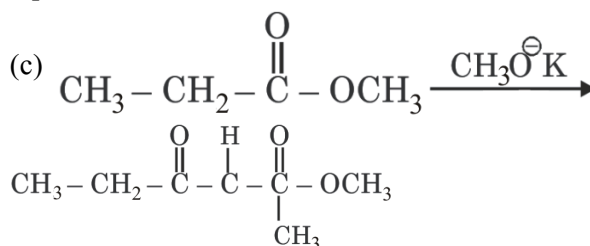
4. Ans (2.00)



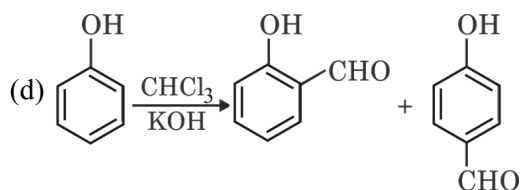
5. Ans (10.00)



4 products



2 products



2 products

6. Ans (7.00)

A, B, C, D, F, G, I

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A)

Let a_n be total ways of division when there are n balls of each colour.

Case I If we make one group $B_1W_1R_1$ remaining balls division can be done in a_{n-1} ways.

Case II If we make one group $B_1W_1R_2$ remaining balls division can be done in a_{n-2} ways.

Case III If we make one group $B_1W_2R_1$ remaining balls division can be done in a_{n-2} ways.

Case IV If we make one group $B_1W_2R_2$ remaining balls division can be done in a_{n-2} ways.

$$\therefore a_n = a_{n-1} + 3a_{n-2}$$

$$\text{Now } a_1 = 1, a_2 = 4$$

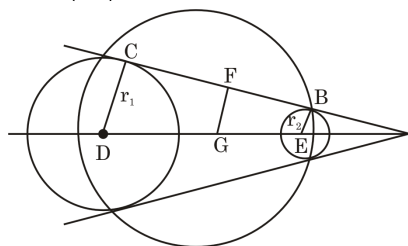
$$\therefore a_3 = 4 + 3 = 7$$

$$a_4 = 7 + 12 = 19$$

$$a_5 = 19 + 21 = 40$$

$$a_6 = 40 + 57 = 97$$

2. Ans (B)



$$FG = \frac{r_1 + r_2}{2}$$

$$GB = \sqrt{(FG)^2 + (FB)^2}$$

$$(GB)^2 = \frac{(r_1 + r_2)^2}{4} + \frac{d^2 - (r_1 - r_2)^2}{4}$$

$$= \frac{d^2 + 4r_1 r_2}{4}$$

$$R_1^2 = \frac{(C_1 C_2)^2 + 4r_1 r_2}{4} = 40$$

$$R_2^2 = \frac{(C_1 C_2)^2 - 4r_1 r_2}{4} = 10$$

3. Ans (C)

Given equation simplifies to

$$(4\sin^2 x - 2\sin^2 z)^2 + 4(2\sin x \cdot \sin z - \sin y)^2 + 2(\sin^2 y - 1)^2 = 0$$

$$\therefore \sin^2 y = 1, 2\sin x \cdot \sin z = \sin y, 2\sin^2 x = \sin^2 z$$

Cases	No. of triplets.
$\sin y = 1,$ $\sin x = \left(\frac{1}{8}\right)^4, \sin z = \left(\frac{1}{2}\right)^4$	4
$\sin y = 1,$ $\sin x = -\left(\frac{1}{8}\right)^4, \sin z = -\left(\frac{1}{2}\right)^4$	4
$\sin y = -1,$ $\sin x = \left(\frac{1}{8}\right)^4, \sin z = -\left(\frac{1}{2}\right)^4$	4
$\sin y = -1,$ $\sin x = -\left(\frac{1}{8}\right)^4, \sin z = +\left(\frac{1}{2}\right)^4$	4

4. Ans (C)

$$\text{Let } f(x) = ax + b$$

$$f(0) = -5 \Rightarrow b = -5$$

$$f(f(0)) = -15 \Rightarrow a = 2$$

$$\therefore f(x) = 2x - 5$$

$$\text{Now } f(x) \cdot f(m-x) \geq 0$$

$$\therefore \left(x - \frac{5}{2}\right) \left(x - \left(m - \frac{5}{2}\right)\right) \leq 0$$

Case I $\frac{5}{2} > m - \frac{5}{2}$, here length of interval
 $= \frac{5}{2} - m + \frac{5}{2} = 2 \Rightarrow m = 3$

Case II $\frac{5}{2} < m - \frac{5}{2}$, here length of interval
 $= m - \frac{5}{2} - \frac{5}{2} = 2$
 $\Rightarrow m = 7$

$\therefore$ two values of m

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,C,D)

$$2A + 3BB^T = I, B^{-1} = A^T \dots\dots (i)$$

$$(2A + 3BB^T)^T = I$$

$$2A^T + 3BB^T = I \dots\dots (ii)$$

$$\therefore A = A^T \text{ also } A^T = B^{-1}$$

$$\therefore A = B^{-1} \text{ or } A^{-1} = B \dots\dots (iii)$$

$$AB = BA = I = BB^{-1}$$

$$\text{Now } A^T = B^{-1}$$

$$A^{-1}B = I$$

$$\text{From (iii) } \therefore B = A^{-1}$$

$$\therefore B^T = (A^{-1})^T = (A^T)^{-1}$$

$$B^T = (B^{-1})^{-1} \therefore B = B^T$$

$$2A + 3B^2 = I \therefore 2AB + 3B^3 = B$$

$$2I + 3B^3 = B \therefore 3B^3 = B - 2I$$

$$\text{Let } \therefore \alpha = |A^{-1} - 3B^3 + BA| = |B - (B - 2I) + I| = |3I| = 27$$

$$\beta = \text{tr}(A^{-1} - 3B^3 - BA) = \text{tr}((B - (B - 2I) - I))$$

$$\beta = \text{tr}(I) = 3$$

6. Ans (A,C,D)

Interchange column twice. Transposing & interchange row twice

$$f(x) = \begin{vmatrix} 5 & 7 & 3 \\ 2 & 9g^2(x) & 2x^2 \\ x & 3g(x) - x^2 & 1 \end{vmatrix} - \begin{vmatrix} 4 & 5 & 2 \\ 2 & 9g^2(x) & 2x^2 \\ x & 3g(x) - x^2 & 1 \end{vmatrix} + \begin{vmatrix} 0 & -2 & 1 \\ 2 & 9g^2(x) & 2x^2 \\ x & 3g(x) - x^2 & 1 \end{vmatrix}$$

$$f(x) = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 9g^2(x) & 2x^2 \\ x & 3g(x) - x^2 & 1 \end{vmatrix} = 9g^2(x) - 6x^2g(x) + 2x^4$$

(i) $m = \int_0^1 [(3g(x) - x^2)^2 + x^4] dx$ for minimum

$$\therefore m = \int_0^1 x^4 dx = \frac{1}{5} \text{ and } g(x) = \frac{x^2}{3}$$

(ii) $\phi(x) = \int_0^x \left(\frac{t^3}{9} - e^t + e^{-t} \right) dt$

$$\phi''(x) = \frac{x^2}{3} - e^x - e^{-x} = 0 \quad \therefore \frac{x^2}{3} = e^x + e^{-x}$$

No solution.

7. Ans (A,C,D)

$$\therefore a_n = 3b_n - 1 \quad \dots (i)$$

$$b_n = a_{n-1} + b_{n-1} + b_{n-1}$$

$$b_n = a_{n-1} + 2b_{n-1} \quad \dots (ii)$$

From (i) and (ii)

$$a_{n+1} = 3a_{n-1} + 2a_n$$

Clearly $a_2 = 0, a_3 = 3$

$$\therefore a_4 = 6$$

$$a_5 = 21$$

$$a_6 = 60$$

$$a_7 = 183$$

8. Ans (A,C)

Three planes intersect in a line

$$\therefore D = D_1 = D_2 = D_3 = 0$$

Here $D_1 = D_2 = D_3 = 0, \therefore D = 0$

$$\Delta = 0$$

$$\begin{vmatrix} -1 & \sin \alpha & \sin \beta \\ \sin \alpha & -1 & \sin \theta \\ \sin \beta & \sin \theta & -1 \end{vmatrix} = 0$$

$$-1(1 - \sin^2 \theta) - \sin \alpha(-\sin \alpha - \sin \theta \sin \beta) + \sin \beta(\sin \alpha \sin \theta + \sin \beta) = 0$$

$$\sin^2 \theta + \sin^2 \alpha + \sin^2 \beta + 2 \sin \alpha \sin \beta \sin \theta = 1$$

Now D.R's of line

$$\begin{vmatrix} i & j & k \\ -1 & \sin \alpha & \sin \beta \\ \sin \alpha & -1 & \sin \theta \end{vmatrix}$$

$$= i(\sin \alpha \sin \theta + \sin \beta) + j(\sin \alpha \sin \beta + \sin \theta) + k(1 - \sin^2 \alpha)$$

This vector is parallel to line.

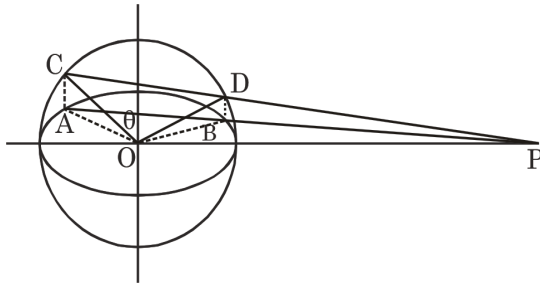
$$\therefore \frac{\sin \alpha \sin \theta + \sin \beta}{\cos \theta} = \frac{\sin \alpha \sin \beta + \sin \theta}{\cos \beta} = \frac{1 - \sin^2 \alpha}{\cos \alpha}$$

$$\sin \alpha \sin \theta + \sin \beta = \cos \theta \cos \alpha$$

$$\cos(\theta + \alpha) = \cos\left(\frac{\pi}{2} - \beta\right)$$

$$\therefore \alpha + \beta + \theta = 2n\pi + \frac{\pi}{2}$$

9. Ans (A,D)



$$A(4 \cos \alpha, 2 \sin \alpha)$$

$$B(4 \cos \beta, 2 \sin \beta)$$

Corresponding points of A and B on auxiliary circle will be

$$C = (4 \cos \alpha, 4 \sin \alpha)$$

$$D = (4 \cos \beta, 4 \sin \beta)$$

$$\text{Area of } \triangle AOB = \frac{1}{2} \text{ ar } (\triangle COD) = \frac{1}{2} \cdot \frac{1}{2} \cdot 4 \cdot \sin \theta$$

$$\therefore \text{Max. area} = 4 \text{ when } \angle COD = \frac{\pi}{2}$$

$$\text{Slope of AB} = \frac{1}{2} \text{ slope of CD}$$

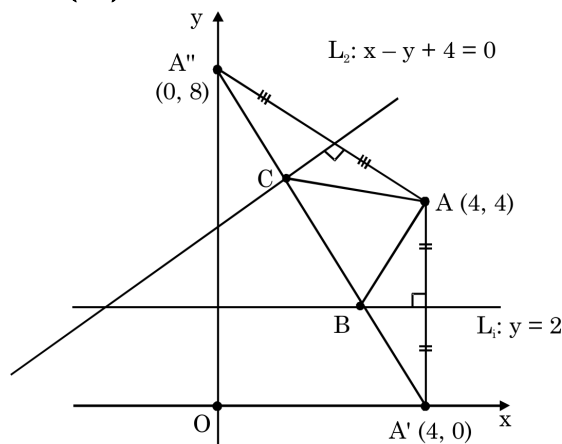
$$\text{Let } m_{CD} = m \therefore \text{CD also through P}$$

$$\text{Equation of CD is } y = m(x - \sqrt{264})$$

$$1 \text{ dist. frm origin} = \frac{|\sqrt{264} \cdot m|}{\sqrt{1 + \tan^2}} = 2\sqrt{2}$$

$$\therefore m = \frac{-1}{4\sqrt{2}}, \quad m_{AB} = \frac{-1}{8\sqrt{2}}$$

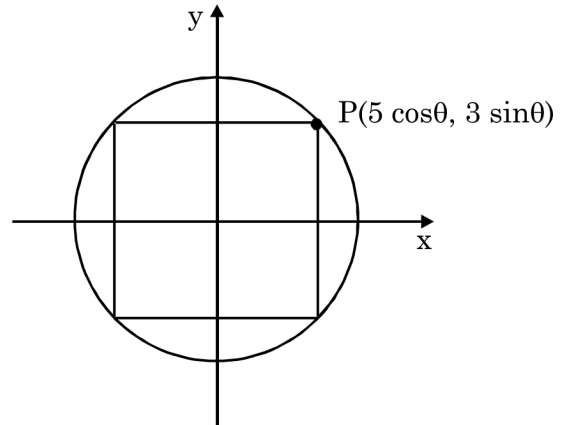
10. Ans (B)



$$\text{Equation of } A'A'' \text{ is } 2x + y = 8$$

$$B \equiv (3, 2) \quad C \equiv \left(\frac{4}{3}, \frac{16}{3}\right)$$

11. Ans (A,B,D)



Area of $R_1 = 10 \cos \theta \cdot 6 \sin \theta = 30 \sin 2\theta$: for this to be maximum

$$\Rightarrow \theta = \pi/4 \Rightarrow P \equiv \left(\frac{5}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$$

Hence, for subsequent areas of rectangles R_n to be maximum the coordinate will be in G.P.

$$\text{with common ratio } r = \frac{1}{\sqrt{2}}$$

$$\Rightarrow a_n = \frac{5}{(\sqrt{2})^{n-1}}; b_n = \frac{3}{(\sqrt{2})^{n-1}}; e_n = \sqrt{1 - \frac{b_n^2}{a_n^2}} = \frac{4}{5}$$

Eccentricity of all the ellipse will be same

Distance between focus of ellipse

$$E_{11} = 2a_{11}e_{11} = 2 \left(\frac{5}{32}\right) \left(\frac{4}{5}\right) = \frac{1}{4}$$

Length of latus rectum of ellipse

$$E_7 = \frac{2b_7^2}{a_7} = \frac{9}{20}$$

$$\therefore \sum_{n=1}^{\infty} (\text{area of } R_n) = 30 + \frac{30}{2} + \frac{30}{2^2} + \dots \text{ upto } \infty = 60$$

$$\Rightarrow \sum_{n=1}^N (\text{area of } R_n) < 60 \text{ for each positive integer } N$$

12. Ans (B,C)

$$Z^4 - |Z|^4 = 12iZ^2$$

$$Z^4 - Z^2 \bar{Z}^2 = 12iZ^2$$

$$Z^2 - \bar{Z}^2 = 12i$$

$$(Z - \bar{Z})(Z + \bar{Z}) = 12i$$

$$\text{Let } Z = x + iy$$

$$2iy(2x) = 12i$$

$$xy = 3$$

Z_1 lies of $xy = 3$ in first quadrant and Z_2 lies on

$xy = 3$ in third quadrant

$$\Rightarrow |Z_1 - Z_2| \text{ is minimum when } Z_1 \equiv (\sqrt{3}, \sqrt{3})$$

$$\text{and } Z_2 \equiv (-\sqrt{3}, -\sqrt{3})$$

$$\Rightarrow |Z_1 - Z_2| \text{ minimum} = 2\sqrt{6}$$

$$\Rightarrow \text{Required area of triangle} = 6$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (7.00)

$$\text{Let } f(x) = ax^3 + bx^2 + cx + d$$

$$f(0) = 5 \Rightarrow d = 5$$

$$f(1) = 8 \Rightarrow a + b + c = 3$$

$$\text{Now, } f'(x) = 3ax^2 + 2bx + c$$

$$f'(0) = 0 \Rightarrow c = 0$$

$$f''(x) = 6ax + 2b$$

$$f''(0) = 0 \Rightarrow b = 0$$

$$\Rightarrow a = 3$$

$$\Rightarrow f(x) = 3x^3 + 5$$

$$\text{Now, } \int_{-1}^1 \frac{(3x^3 + 5) dx}{(1 + x^2)} = 10 [\tan^{-1}(x)]_0^1$$

$$= 10 \frac{\pi}{4} = \frac{5\pi}{2}$$

$$m = 5, n = 2$$

$$m + n = 7$$

2. Ans (11)

$$f(x) = x^3 + (x - 1)^3 - 4x + \frac{5}{2}$$

$$f(1 - x) = (1 - x)^3 + (-x)^3 - 4(1 - x) + \frac{5}{2}$$

$$\Rightarrow f(x) + f(1 - x) = 1$$

$$\text{Let } I = \int_{\frac{1}{8}}^{\frac{7}{8}} f(f(x)) dx \dots\dots\dots (1)$$

Apply king property

$$I = \int_{\frac{1}{8}}^{\frac{7}{8}} f(1 - f(x)) dx \dots\dots\dots (2)$$

Adding (1) and (2)

$$2I = \int_{\frac{1}{8}}^{\frac{7}{8}} f(f(x)) dx + \int_{\frac{1}{8}}^{\frac{7}{8}} f(1 - f(x)) dx$$

$$2I = \int_{\frac{1}{8}}^{\frac{7}{8}} dx = \frac{3}{4}$$

$$\Rightarrow I = \frac{3}{8}$$

$$m = 3$$

$$n = 8$$

$$m + n = 11$$

3. Ans (11.00)

$$\text{Let } I = \int_2^4 x h(x) dx$$

By parts,

$$I = (x f^{-1}(x))_2^4 - \int_2^4 f^{-1}(x) dx$$

$$I = 4f^{-1} - 2f^{-1}(2) - \int_2^4 f^{-1}(x) dx$$

$$f^{-1}(4) = 1$$

$$f^{-1}(2) = 0$$

We know,

$$\int_a^b f(x) dx + \int_{f(a)}^{f(b)} f^{-1}(x) dx = bf(b) - af(a)$$

$$\int_0^1 f(x) dx + \int_2^4 f^{-1}(x) dx = 1 \cdot f(1) - 0 \cdot f(0)$$

$$\text{Now, } \int_0^1 f(x) dx = \int_0^1 (x^3 + x + \sin 2\pi x + 2) dx$$

$$= \left(\frac{x^4}{4} + \frac{x^2}{2} - \frac{\cos 2\pi x}{2\pi} + 2x \right)_0^1$$

$$= \left(\frac{1}{4} + \frac{1}{2} - \frac{1}{2\pi} + 2 \right) - \left(-\frac{1}{2\pi} \right) = \frac{11}{4}$$

$$\text{So, } \frac{11}{4} + \int_2^4 f^{-1}(x) dx = 4$$

$$\Rightarrow \int_2^4 f^{-1}(x) dx = \frac{5}{4}$$

$$\text{Now, } I = 4 - \frac{5}{4} = \frac{11}{4}$$

4. Ans (4.00)

Let

$$\frac{P(\text{a prime number})}{2} = \frac{P(\text{a composite})}{1} = \frac{P(1)}{3} = k$$

$$\text{So, } P(\text{a prime number}) = 2k$$

$$P(\text{a composite number}) = k$$

$$\& P(1) = 3k$$

$$\& 3 \times 2k + 2 \times k + 3k = 1$$

$$\Rightarrow k = \frac{1}{11}$$

$$P(\text{success}) = P(1 \text{ or } 4) = 3k + k = \frac{4}{11}$$

$$\text{Number of trials, } n = 11$$

$$\text{Mean} = np = 11 \times \frac{4}{11} = 4$$

5. Ans (10.00)

The given sequence 4, 7, 1, 8, 9, 7, 6, 3, 9, 2, 1, 3, 4, 7, 1

$$S_{12} = 4 + 7 + 1 + \dots + 1 + 3 = 60$$

$$S_n > 10004$$

$$\frac{10004}{60} = 166 \text{ and } 166 \times 12 = 1992$$

$$S_{1992} = 60 \times 166 = 9960$$

$$9960 + 4 + 7 + 1 + 8 + 9 + 7 + 6 = 10002 + 3 = 10005$$

$$\text{No. of terms} = 2000$$

$$S_{2000} > 10004$$

6. Ans (1.00)

$$N^r : {}^nC_0 \cdot {}^{n+k-1}C_{n-1} + {}^nC_1 \cdot {}^{n+k-2}C_{n-1} +$$

$${}^nC_2 \cdot {}^{n+k-3}C_{n-1} + \dots + {}^nC_{k-1} \cdot {}^nC_{n-1} + {}^nC_k \cdot {}^{n-1}C_{n-1}$$

$$= \text{coefficient of } x^{n-1} \text{ in}$$

$${}^nC_0 (1+x)^{n+k-1} + {}^nC_1 (1+x)^{n+k-2} + \dots +$$

$${}^nC_k (1+x)^{n-1} + {}^nC_{k+1} (1+x)^{n-2} + \dots + {}^nC_n (1+x)^{k-1}$$

$$= \text{coefficient of } x^{n-1} \text{ in}$$

$$(1+x)^{n+k-1} (1+(1+x)^{-1})^n = (1+x)^{k-1} (x+2)^n$$

$$= \sum {}^nC_r 2^r x^{n-r} (1+x)^{k-1}$$

$$= \sum_{r=1}^n 2^r \cdot {}^nC_r \cdot {}^{k-1}C_{r-1}$$

$$\boxed{N^r = D^r}$$

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B	D	A,C	A,B	A,C,D	A,B,C	A	A,B,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C	D	A	B				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	168.00	31.00	1.00	50.00	3.00	3.00		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	B,D	A,B,C	B,C	A,C,D	A,C	B,D	A,C	A,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	B	C	C	D				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	-0.18 to -0.19	83.33	120.00	4.00	9.74	11.00		

PART-3 : MATHEMATICS

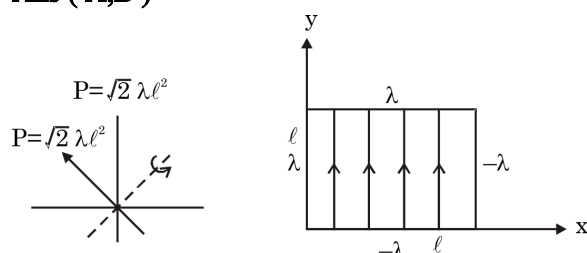
SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	C,D	A,B,C	A,C	A,B,C	A,B,C	A,C	B,C,D	C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C	D	B	C				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	2.00	37.00	5.00	6.00	2.00	4.00		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B)



$$\sqrt{2}\lambda l^2 E = \frac{1}{2} I w^2$$

$$\sqrt{2}\lambda l^2 E = \frac{1}{2} \left(\frac{m l^2}{3} \sin^2 45 \right) \times 4 w^2$$

$$\sqrt{2}\lambda l^2 E = \frac{m l^2}{3} w^2 \quad \text{motion will be oscillatory.}$$

$$w = \sqrt{\frac{3\sqrt{2}\lambda E}{m}} \Rightarrow w = \sqrt{\frac{3\sqrt{2} \times 1 \times 4}{1}} = \sqrt{12\sqrt{2}}$$

2. Ans (D)

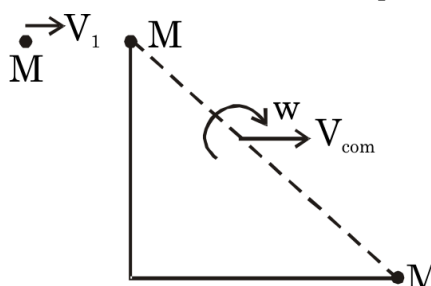
$$\frac{E}{500 + R_g} = C\theta \quad \dots (1)$$

$$\left[\frac{E}{500 + \frac{R_g \times 10}{R_g + 10}} \right] \frac{10}{(10 + R_g)} = \frac{C\theta}{28} \quad \dots (2)$$

From (1) & (2) $R_g = 100 \Omega$

3. Ans (A,C)

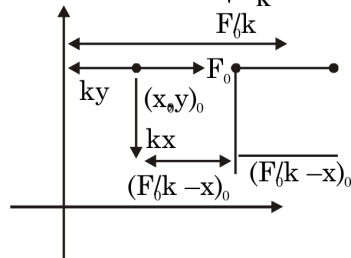
Applying conservation of momentum and angular momentum about com of L-shape frame.



5. Ans (A,C,D)

$$x = \left(\frac{F_0}{k} - x_0 \right) + x_0 = \left(\frac{2F_0}{k} - x_0 \right)$$

$$y = -y_0 \Rightarrow T = 2\pi\sqrt{\frac{m}{k}}$$

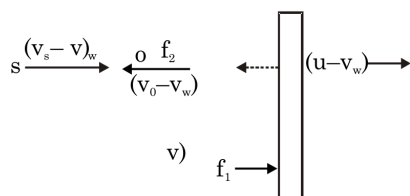


7. Ans (A)

$$v_s = 20 \text{ m/s}, v_0 = 10 \text{ m/s}, u = 5 \text{ m/s}, v_w = 30 \text{ m/s},$$

$$f_0 = 500 \text{ Hz}, v = 300 \text{ m/s}$$

w.r.t. wind



$$f_1 = \frac{v - (u - v_w)}{v - (v_s - v_w)} f_0, f_2 = \left(\frac{v + v_0 - v_w}{v + u - v_w} \right) f_1,$$

$$f_2 = \frac{(v + v_0 - v_w)(v + v_w - u)}{(v + u - v_w)(v + v_w - v_s)} f_0$$

$$f_2 = \frac{(300 - 30 + 10)(300 + 30 - 5)}{(300 + 30 - 20)(300 - 30 + 5)} f_0$$

$$= \frac{280 \times 325}{310 \times 275} \times 500$$

$$= \frac{28}{31} \times \frac{65}{55} \times 500 = \frac{28 \times 13}{31 \times 11} \times 500 = \frac{364}{341} \times 500 = 534 \text{ Hz}$$

Wavelength of sound received by wall

$$\lambda = \frac{v - (v_s - v_w)}{f_0} = \frac{v + v_w - v_s}{f_0}$$

$$= \frac{300 + 30 - 20}{500} = 0.62 \text{ m}$$

Wavelength of sound received by listener after reflection from wall

$$\frac{v - u - v_w}{f_1} = \frac{(v + u - v_w)(v + v_w - v_s)}{(v + v_w - u) f_0}$$

$$= \frac{(300 + 5 - 30)(300 + 30 - 20)}{(300 + 30 - 5) \times 500}$$

$$= \frac{275 \times 310}{325 \times 500} = 0.52 \text{ m}$$

8. Ans (A,B,C)

For just reaching at D, particle just complete

lower circular motion.

$$(A) mgh = \frac{1}{2} m(5gR) \Rightarrow h = \frac{5R}{2}$$

$$(B) \text{ Velocity at C : } mg(h - 2R) = \frac{1}{2} mV_c^2$$

$$\Rightarrow V_c = \sqrt{gR}$$

$$N = mg + \frac{mV_c^2}{R/16} = 17mg$$

(C) Energy conservation from C to D :-

$$\frac{1}{2} m(gR) = mg \cdot \frac{R}{8} + \frac{1}{2} mV_D^2$$

$$V_D = \sqrt{\frac{3gR}{4}}$$

time of flight :

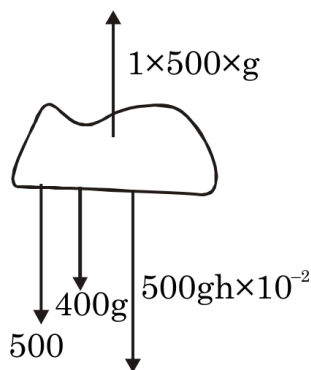
$$\sqrt{\frac{2h}{g}} = \sqrt{\frac{17R}{4g}}$$

$$\text{Distance} = V_D T = \sqrt{\frac{51}{16}} R$$

PART-1 : PHYSICS

SECTION-I (ii)

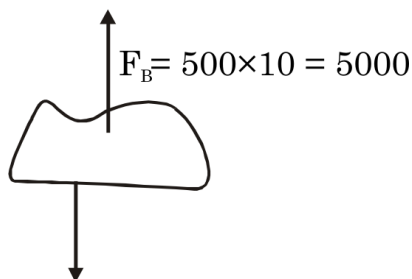
9. Ans (C)



$$500g = 500 + 500gh \times 10^{-2} + 4000$$

$$500 = 50h \Rightarrow h = 10 \text{ m}$$

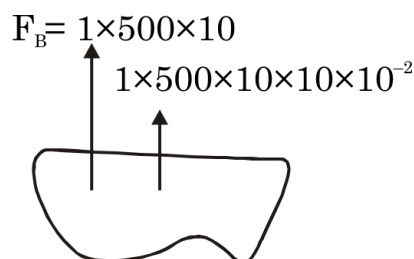
Force by liquid on upper object



$$500gh \times 10^{-2} = 500 \text{ N}$$

$$5000 - 500 = 4500 \text{ N}$$

Force by liquid on lower object

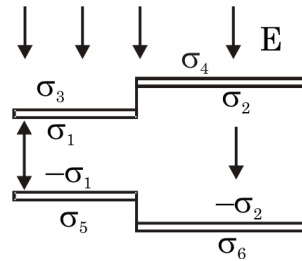


$$5000 + 500 = 5500$$

Total force by liquid on both object

$$5500 + 4500 = 10^4 \text{ N}$$

11. Ans (A)



$$\sigma_1 A_1 + \sigma_2 A_2 = -(\sigma_3 A_1 + \sigma_4 A_2)$$

$$\sigma_1 A_1 + \sigma_2 A_2 = (\sigma_5 A_1 + \sigma_6 A_2)$$

$$2(\sigma_1 A_1 + \sigma_2 A_2) = (\sigma_5 - \sigma_3) A_1 + (\sigma_6 - \sigma_4) A_2$$

$$\frac{\sigma_1}{\epsilon_0} = E_1, \frac{\sigma_2}{\epsilon_0} = E_2, \frac{(\sigma_5 - \sigma_3)}{2\epsilon_0} = E, \frac{(\sigma_6 - \sigma_4)}{2\epsilon_0} = E$$

$$2A_1 E_1 \epsilon_0 + 2A_2 E_2 \epsilon_0 = 2\epsilon_0 E A_1 + 2\epsilon_0 E A_2$$

$$A_1 E_1 + A_2 E_2 = E(A_1 + A_2) \quad \dots(1)$$

$$E_1 d_1 = E_2 d_2 \quad \dots(2)$$

$$E_1 = E d_2 \frac{(A_1 + A_2)}{A_1 d_2 + A_2 d_1} = \frac{100 \times 2(1+2)}{(2+2 \times 1)} = \frac{200 \times 3}{4} = 150$$

$$E_2 = E d_1 \frac{(A_1 + A_2)}{(A_1 d_2 + A_2 d_1)} = 75$$

$$E_1 + E_2 = 225$$

$$E_1 - E_2 = 75$$

PART-1 : PHYSICS

SECTION-II

1. Ans (168.00)

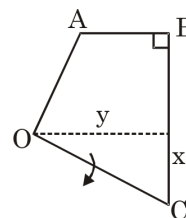
$$x + y = 10$$

$$y - x = 2$$

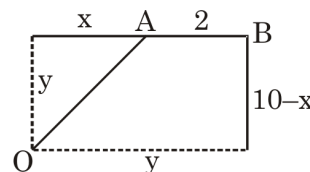
$$2y = 12$$

$$y = 6 \text{ cm}$$

$$x = 4 \text{ cm}$$



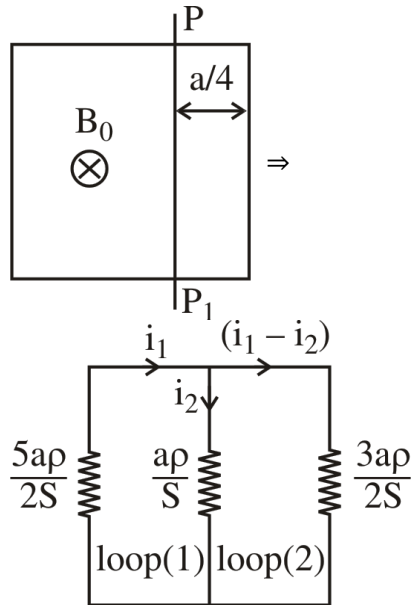
It becomes a square of side length = 6 cm



$$I = \frac{2}{3} m l^2 = \frac{2}{3} \times 7 \times (6)^2$$

$$= 14 \times 12 = 168 \text{ kg cm}^2$$

2. Ans (31.00)



In loop (1) :

$$-i_1 \frac{5a\rho}{2S} - i_2 \frac{a\rho}{S} = \frac{dB}{dt} \times \frac{3a^2}{4} \quad \dots(1)$$

in loop (2) :

$$-(i_1 - i_2) \frac{3a\rho}{2S} + i_2 \frac{a\rho}{S} = \frac{dB}{dt} \times \frac{a^2}{4} \quad \dots(2)$$

From equation (1) & (2)

$$i_2 = -\frac{2}{31} \times \frac{dB}{dt} \times \frac{aS}{\rho}$$

Force on jumper wire :

$$F = \frac{dp}{dt} = i\ell B$$

$$\frac{dp}{dt} = i_2 a B$$

$$\frac{dp}{dt} = -\frac{2}{31} \frac{dB}{dt} \times \frac{a^2 S B}{\rho}$$

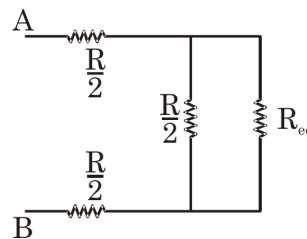
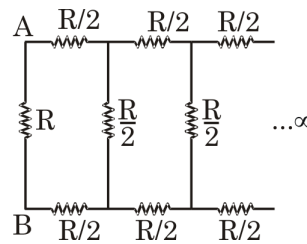
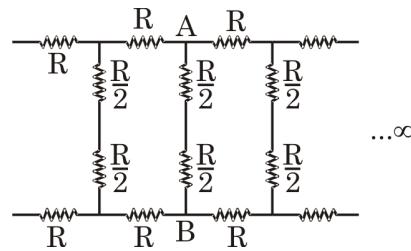
$$\int_0^{mv} dP = -\frac{2}{31} \frac{a^2 S}{\rho} \int_{B_0}^0 B dB$$

$$v = \frac{a^2 B^2 S}{31 \rho m}$$

$$\alpha = 31$$

3. Ans (1.00)

Using mirror symmetry drawing in two D plane.



$$\frac{R_{eq} \frac{R}{2}}{R_{eq} + \frac{R}{2}} + R = R_{eq}$$

$$\frac{R R_{eq}}{2R_{eq} + R} = (R_{eq} - R)$$

$$\Rightarrow 2R_{eq}^2 + R_{eq}R - 2RR_{eq}$$

$$-R^2 = RR_{eq}$$

$$2R_{eq}^2 - 2RR_{eq} - R^2 = 0$$

$$R_{eq} = \frac{2R \pm \sqrt{4R^2 + 8R^2}}{4} = \frac{2R \pm 2\sqrt{3}R}{4} = \frac{R + \sqrt{3}R}{2}$$

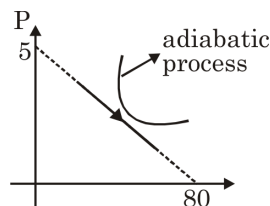
$$R_{AB} = \frac{1}{R} + \frac{2}{(R + \sqrt{3}R)} = \frac{3R + \sqrt{3}R}{R^2(1 + \sqrt{3})} = \frac{\sqrt{3}R(1 + \sqrt{3})}{R^2(1 + \sqrt{3})}$$

$$\Rightarrow R_{AB} = \left(\frac{R}{\sqrt{3}} \right) = \frac{\sqrt{3}}{\sqrt{3}} = 1 \Omega$$

4. Ans (50.00)

$$P = -\frac{1}{16}V + 5 \quad \dots(i)$$

$$PV^\gamma = \text{const.}$$



$$\text{Slope} = -\gamma \frac{P}{V}$$

$$-\gamma \frac{P}{V} = -\frac{1}{16}$$

$$P = \frac{V}{16V} \quad \dots(ii)$$

Equation (i) and (ii)

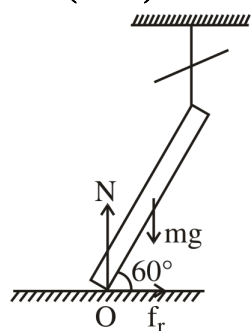
$$\frac{V}{16 \times \gamma} = -\frac{V}{16} + 5$$

$$\frac{V \times 3}{16 \times 5} + \frac{V}{16} = 5$$

$$\frac{V}{16} \times \frac{8}{5} = 5$$

$$V = 50 \text{ litre}$$

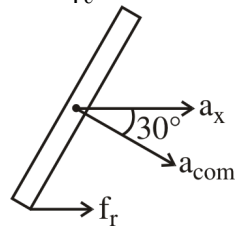
5. Ans (3.00)



Torque about point O,

$$mg \cdot \frac{l}{2} \cos 60^\circ = \frac{ml^2}{3} \times \alpha$$

$$\alpha = \frac{3g}{4l}$$



$$a_{\text{com}} = \alpha \cdot \frac{l}{2} = \frac{3g}{8}$$

$$a_x = a_{\text{com}} \cdot \cos 30^\circ = \frac{3\sqrt{3}}{16}g$$

$$f_r = ma_x = mg \frac{3\sqrt{3}}{16}$$

$$f_r = 3$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (B,D)

The electron pair back donated by metal goes into antibonding MO of carbon monoxide, thus decreases the bond order and increases the bond length.

The π -acceptor tendency of ligands are in order $\text{PF}_3 > \text{PCl}_3 > \text{PMe}_3$.

2. Ans (A,B,C)

Magnesium metal can't be extracted from aqueous solution.

5. Ans (A,C)

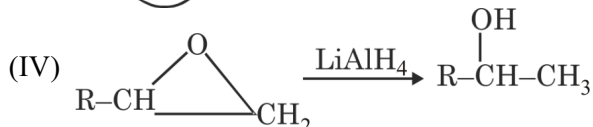
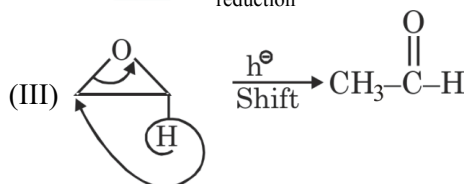
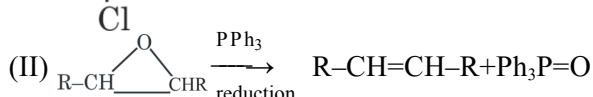
$$\Delta T_b = 0.54 \times \frac{10}{540} \times 1000 = 10$$

$$P^s = P^\circ X_{\text{H}_2\text{O}} = 35 \times \frac{30}{40} = 26.25$$

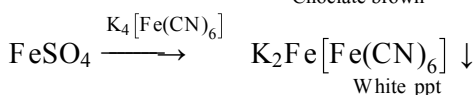
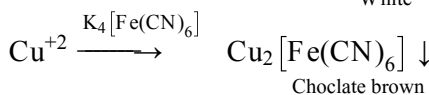
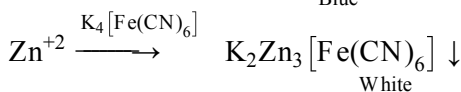
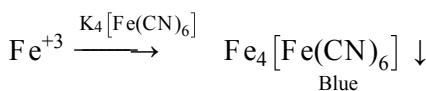
PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (B)



12. Ans (D)



PART-2 : CHEMISTRY

SECTION-II

1. Ans (-0.18 to -0.19)

$$\Delta H_{\text{reaction}}^0 = 29.0 \text{ kJ}$$

$$\Delta S_{\text{reaction}}^0 = 95.1 \text{ JK}^{-1}$$

$$\Delta G_{\text{reaction}}^0 = \Delta H^0 - T \Delta S^0$$

$$= 29000 - 300 \times 95.1$$

$$= 470 \text{ J}$$

$$\ln K = \frac{-\Delta G^0}{RT} = -0.188$$

2. Ans (83.33)

$$\text{No. of m mol. of FeC}_2\text{O}_4 = 100 \times \frac{1}{20} = 5 \text{ m mol.}$$

$$\text{The solution now has } \frac{5}{3} + 2y \text{ mol of}$$

Fe⁺³ Which will convert to Fe⁺²

$$\therefore \frac{5}{3} + 2y = 10$$

Mass of Fe₂(SO₄)₃

$$= \frac{25}{6} \times 400 = \frac{5000}{3} \text{ mg} = \frac{5}{3} \text{ g}$$

Mass of FeC₂O₄ = 0.24 g

$$\% \text{ purity of Fe}_2(\text{SO}_4)_3 = \frac{5/3}{2} \times 100 = 83.33$$

3. Ans (120.00)

For rate to reach 36%

$$A_t = 0.6 \text{ \AA}$$

$$t = \frac{1}{k} \left[\frac{1}{0.6 \text{ \AA}} - \frac{1}{\text{A}} \right]$$

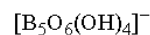
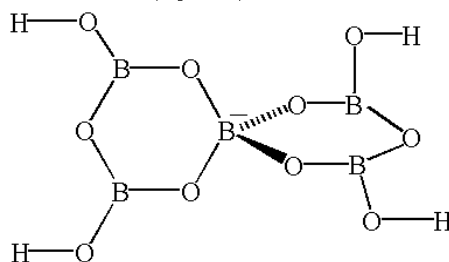
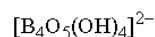
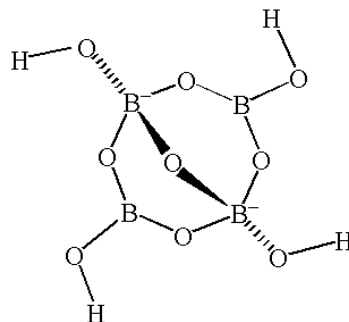
$$t = \frac{1}{k} \left[\frac{2}{3 \text{ \AA}} \right]$$

$$k = \frac{1}{1440 \times 7}$$

$$\therefore t = 1440 \times 7 \left[\frac{2}{3} \times \frac{1}{56 \times 10^{-3}} \right]$$

$$t = 120 \times 10^3 \text{ sec}$$

6. Ans (11.00)



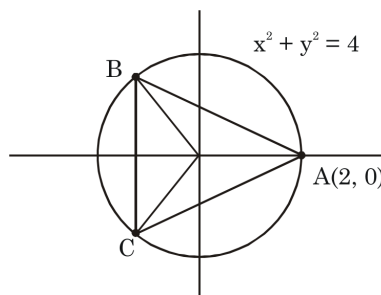
PART-3 : MATHEMATICS

SECTION-I (i)

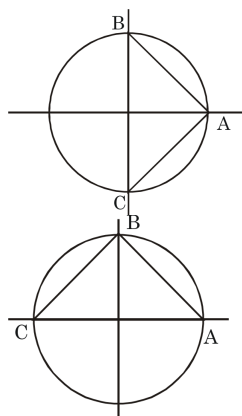
1. Ans (C,D)

For equilateral triangle (0, 0) is centroid and circumcentre

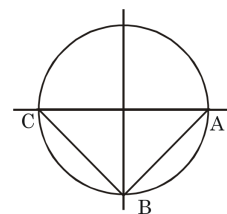
i.e. m = 1



For right angled isosceles



n = 3.



2. Ans (A,B,C)

$$h(t) = \int_{-1}^1 f\left(\frac{x}{t}\right) dx; x = ty$$

$$= \int_{-\frac{1}{t}}^{\frac{1}{t}} f(y) \cdot t dy; \frac{h(t)}{t} = \int_{-\frac{1}{t}}^{\frac{1}{t}} 3^{-|y|} \cdot dy;$$

$$= 2 \int_0^{\frac{1}{t}} 3^{-|y|} \cdot dy$$

$$\text{Now LHL} = 2 \int_0^{\infty} 3^{-y} \cdot dy = \frac{2}{\ln 3} [3^{-y}]_0^{\infty} = -\frac{2}{\ln 3}$$

$$\text{RHL} = 2 \int_0^{\infty} 3^{-y} \cdot dy = -\frac{2}{\ln 3} [3^{-y}]_0^{\infty} = \frac{2}{\ln 3}$$

3. Ans (A,C)

$$g(x+1) = \frac{1}{3} g(x);$$

$$g(x+2) = \frac{1}{3} g(x+1) = \frac{1}{9} g(x)$$

Now,

$$\lim_{n \rightarrow \infty} \alpha_n = \lim_{n \rightarrow \infty} \left(\int_0^1 g(x) dx + \int_1^2 g(x) dx + \int_2^3 g(x) dx + \dots \right)$$

$$\int_2^3 g(x) dx; x = t + 2; \int_0^1 g(t+2) dt = \frac{1}{9} \int_0^1 g(x) dx$$

$$\text{i.e. } \lim_{n \rightarrow \infty} \alpha_n = \int_0^1 g(x) dx \left(1 + \frac{1}{3} + \frac{1}{9} + \dots \right)$$

$$= \frac{3}{2} \int_0^1 g(x) dx$$

4. Ans (A,B,C)

$$x \in (0, \cos 1) \Rightarrow \cos^{-1} x > 1 > \sin^{-1} x$$

$$x \in \left(\cos 1, \frac{1}{\sqrt{2}} \right) \Rightarrow \sin^{-1} x < \cos^{-1} x < 1$$

$$x \in (\sin 1, 1) \Rightarrow \sin^{-1} x > 1 > \cos^{-1} x$$

$$x \in \left(\frac{1}{\sqrt{2}}, \sin 1 \right) \Rightarrow \cos^{-1} x < \sin^{-1} x < 1$$

5. Ans (A,B,C)

$$C_2 = 4C_1; C_3 = 4C_2 = 16C_1$$

$$R_2 = 6 + R_1; R_3 = 6 + R_2 = 12 + R_1$$

$$R_1 + R_2 + R_3 = C_1 + C_2 + C_3$$

$$\Rightarrow 18 + 3R_1 = 21C_1$$

$$\Rightarrow 6 + R_1 = 7C_1 = R_2$$

$$M = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 7 \\ 1 & 4 & 8 \end{bmatrix}$$

6. Ans (A,C)

$$\sum_{j=1}^{200} \sum_{k=1}^{200} \frac{Z_j}{Z_k} = (Z_1 + Z_2 + \dots + Z_{200})$$

$$\left(\frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_{200}} \right) \dots (I)$$

$$\sum_{j=1}^{200} \sum_{k=1}^{200} \frac{Z_j}{Z_k} = 0$$

7. Ans (B,C,D)

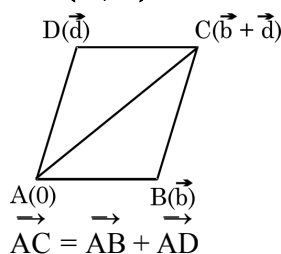
$$D = \begin{vmatrix} 1 + \ell_1 m_1 + \ell_1^2 m_1^2 & 1 + \ell_2 m_1 + \ell_2^2 m_1^2 & 1 + \ell_3 m_1 + \ell_3^2 m_1^2 \\ 1 + \ell_1 m_2 + \ell_1^2 m_2^2 & 1 + \ell_2 m_2 + \ell_2^2 m_2^2 & 1 + \ell_3 m_2 + \ell_3^2 m_2^2 \\ 1 + \ell_1 m_3 + \ell_1^2 m_3^2 & 1 + \ell_2 m_3 + \ell_2^2 m_3^2 & 1 + \ell_3 m_3 + \ell_3^2 m_3^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & \ell_1 & \ell_1^2 \\ 1 & \ell_2 & \ell_2^2 \\ 1 & \ell_3 & \ell_3^2 \end{vmatrix} \begin{vmatrix} 1 & m_1 & m_1^2 \\ 1 & m_2 & m_2^2 \\ 1 & m_3 & m_3^2 \end{vmatrix}$$

$$= (\ell_1 - \ell_2)(\ell_2 - \ell_3)(\ell_3 - \ell_1)(m_1 - m_2)(m_2 - m_3)(m_3 - m_1)$$

$$= +ve \text{ so } D > 0$$

8. Ans (C,D)



$$\frac{2}{\gamma} \vec{AQ} = \frac{\vec{AP}}{\alpha} + \frac{\vec{AR}}{\beta}$$

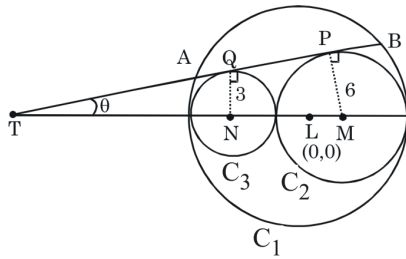
$$\Rightarrow \frac{\vec{AP}}{\alpha} + \left(-\frac{2}{\gamma} \right) \vec{AQ} + \frac{\vec{AR}}{\beta} = 0$$

$$\Rightarrow \frac{1}{\alpha} - \frac{2}{\gamma} + \frac{1}{\beta} = 0 \Rightarrow \frac{1}{\alpha} + \frac{1}{\beta} = \frac{2}{\gamma}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

9. Ans (C)



$$\text{Length of } PQ = TCT = \sqrt{9^2 - (6-3)^2} = 6\sqrt{2}$$

$$\text{let } C_1 : x^2 + y^2 = 9^2$$

$$\text{let } \angle QTN = \theta$$

$$\therefore \frac{NT}{MT} = \frac{r_3}{r_2} = \frac{3}{6} = \frac{1}{2} = \frac{TQ}{TP} \Rightarrow N \text{ is mid-point of } TM$$

$$\Rightarrow TP = 2TQ \Rightarrow TQ = PQ$$

$$\Rightarrow TQ = 6\sqrt{2}$$

$$\therefore NT = \sqrt{72+9} = 9$$

$$\Rightarrow LT = 9 + (9-3) = 15$$

$$\therefore L(0,0) \Rightarrow T(-15,0)$$

$$\sin \theta = \frac{NQ}{NT} = \frac{3}{9} = \frac{1}{3}$$

$$\Rightarrow \cos \theta = \frac{6\sqrt{2}}{9} = \frac{2\sqrt{2}}{3}$$

$$\text{Any point on chord } AB \text{ be } \left(-15 + \frac{2\sqrt{2}r}{3}, \frac{r}{3}\right)$$

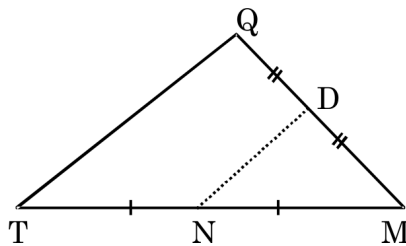
$$\text{lie on circle } C_1 : x^2 + y^2 = 81$$

$$\Rightarrow r^2 - 20\sqrt{2}r + 144 = 0 \begin{cases} r_1 \\ r_2 \end{cases}$$

$$\therefore AB = |r_1 - r_2| = \sqrt{(r_1 + r_2)^2 - 4r_1r_2}$$

$$= \sqrt{800 - 576} = \sqrt{224}$$

$$= 4\sqrt{14}$$



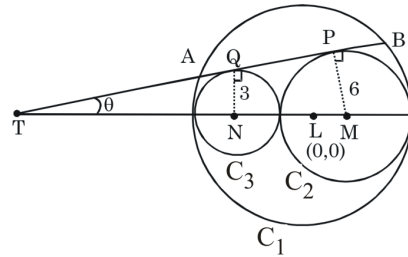
$\therefore N$ is mid-point of TM and D is mid-point of QM

$$\Rightarrow ND = \frac{1}{2}TQ$$

$$= \frac{1}{2}(6\sqrt{2})$$

$$= 3\sqrt{2}$$

10. Ans (D)



$$\text{Length of } PQ = TCT = \sqrt{9^2 - (6-3)^2} = 6\sqrt{2}$$

$$\text{let } C_1 : x^2 + y^2 = 9^2$$

$$\text{let } \angle QTN = \theta$$

$$\therefore \frac{NT}{MT} = \frac{r_3}{r_2} = \frac{3}{6} = \frac{1}{2} = \frac{TQ}{TP} \Rightarrow N \text{ is mid-point of } TM$$

$$\Rightarrow TP = 2TQ \Rightarrow TQ = PQ$$

$$\Rightarrow TQ = 6\sqrt{2}$$

$$\therefore NT = \sqrt{72+9} = 9$$

$$\Rightarrow LT = 9 + (9-3) = 15$$

$$\therefore L(0,0) \Rightarrow T(-15,0)$$

$$\sin \theta = \frac{NQ}{NT} = \frac{3}{9} = \frac{1}{3}$$

$$\Rightarrow \cos \theta = \frac{6\sqrt{2}}{9} = \frac{2\sqrt{2}}{3}$$

$$\text{Any point on chord } AB \text{ be } \left(-15 + \frac{2\sqrt{2}r}{3}, \frac{r}{3}\right)$$

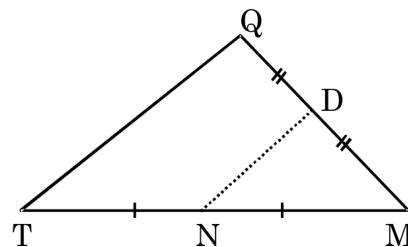
$$\text{lie on circle } C_1 : x^2 + y^2 = 81$$

$$\Rightarrow r^2 - 20\sqrt{2}r + 144 = 0 \begin{cases} r_1 \\ r_2 \end{cases}$$

$$\therefore AB = |r_1 - r_2| = \sqrt{(r_1 + r_2)^2 - 4r_1r_2}$$

$$= \sqrt{800 - 576} = \sqrt{224}$$

$$= 4\sqrt{14}$$



$\therefore N$ is mid-point of TM and D is mid-point of QM

$$\Rightarrow ND = \frac{1}{2}TQ$$

$$= \frac{1}{2}(6\sqrt{2})$$

$$= 3\sqrt{2}$$

11. Ans (B)

$$f(x) = \begin{cases} -2; & -1 \leq x < -\frac{1}{2} \\ -1; & -\frac{1}{2} \leq x < 0 \\ 0; & 0 \leq x < \frac{1}{2} \\ 1; & \frac{1}{2} \leq x < 1 \\ 2x; & 1 \leq x \leq 2 \end{cases}$$

$$f(g(x)) = \begin{cases} -2; & x \in \left[\frac{7\pi}{6}, \frac{11\pi}{6}\right] - \left\{\frac{3\pi}{2}\right\} \\ -1; & x = \frac{3\pi}{2} \\ 1; & x \in \left(\frac{\pi}{6}, \frac{5\pi}{6}\right) \\ \operatorname{cosec} x; & x \in \left(0, \frac{\pi}{6}\right] \cup \left[\frac{5\pi}{6}, \pi\right) \end{cases}$$

$$f(f(x)) = \begin{cases} -2; & -\frac{1}{2} \leq x < 0 \\ 0; & 0 \leq x < \frac{1}{2} \\ 2; & \frac{1}{2} \leq x < 1 \\ 4; & x = 1 \end{cases}$$

$$g(f(x)) = \begin{cases} \frac{\operatorname{cosec} 1}{2}; & \frac{1}{2} \leq x < 1 \\ \frac{\operatorname{cosec} 2x}{2}; & 1 \leq x \leq 2 - \left\{\frac{\pi}{2}\right\} \end{cases}$$

$$f(2g(x)) = \begin{cases} -2; & x = \frac{3\pi}{2} \\ 2\operatorname{cosec} x; & x \in \left[\frac{\pi}{6}, \frac{5\pi}{6}\right] \end{cases}$$

12. Ans (C)

$$f(x) = \begin{cases} -2; & -1 \leq x < -\frac{1}{2} \\ -1; & -\frac{1}{2} \leq x < 0 \\ 0; & 0 \leq x < \frac{1}{2} \\ 1; & \frac{1}{2} \leq x < 1 \\ 2x; & 1 \leq x \leq 2 \end{cases}$$

$$f(g(x)) = \begin{cases} -2; & x \in \left[\frac{7\pi}{6}, \frac{11\pi}{6}\right] - \left\{\frac{3\pi}{2}\right\} \\ -1; & x = \frac{3\pi}{2} \\ 1; & x \in \left(\frac{\pi}{6}, \frac{5\pi}{6}\right) \\ \operatorname{cosec} x; & x \in \left(0, \frac{\pi}{6}\right] \cup \left[\frac{5\pi}{6}, \pi\right) \end{cases}$$

$$f(f(x)) = \begin{cases} -2; & -\frac{1}{2} \leq x < 0 \\ 0; & 0 \leq x < \frac{1}{2} \\ 2; & \frac{1}{2} \leq x < 1 \\ 4; & x = 1 \end{cases}$$

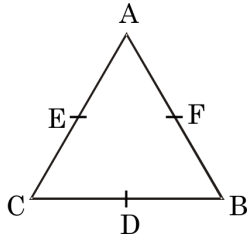
$$g(f(x)) = \begin{cases} \frac{\operatorname{cosec} 1}{2}; & \frac{1}{2} \leq x < 1 \\ \frac{\operatorname{cosec} 2x}{2}; & 1 \leq x \leq 2 - \left\{\frac{\pi}{2}\right\} \end{cases}$$

$$f(2g(x)) = \begin{cases} -2; & x = \frac{3\pi}{2} \\ 2\operatorname{cosec} x; & x \in \left[\frac{\pi}{6}, \frac{5\pi}{6}\right] \end{cases}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (2.00)



$$D(a\lambda, 0, 0) \quad E(0, b\lambda, 0) \quad F(0, 0, c\lambda)$$

$$p = 2EF \quad \Rightarrow p^2 = 4(EF)^2$$

$$\Rightarrow p^2 = 4(b^2 + c^2)\lambda^2$$

$$q^2 = 4(a^2 + c^2)\lambda^2$$

$$r^2 = 4(a^2 + b^2)\lambda^2$$

$$p^2 + q^2 + r^2 = 8\lambda^2 (a^2 + b^2 + c^2)$$

$$p^2 + q^2 + r^2 = 8\lambda^2 \left(\frac{p^2 + q^2 + r^2}{2} \right) \Rightarrow \lambda^2 = 1/4$$

2. Ans (37.00)

(1,1), (1,4), (4,1), (2,4), (4,2), (3,4), (4,3), (4,4) – all have one choice for image.

(2,1), (1,2), (2,2) – all have three choice for image

(3,2), (2,3), (3,1), (1,3), (3,3) – all have two choices for image.

So the total functions = $3 \times 3 \times 2 \times 2 \times 2 = 72$

Case 1 : None of the pre-images have 3 as image

Total functions = $2 \times 2 \times 1 \times 1 \times 1 = 4$

Case 2 : None of the pre-images have 2 as image

Total functions = $2 \times 2 \times 2 \times 2 \times 2 = 32$

Case 3 : None of the pre-images have either 3 or 2 as image

Total functions = $1 \times 1 \times 1 \times 1 \times 1 = 1$

$\therefore$ Total onto functions = $72 - 4 - 32 + 1 = 37$

3. Ans (5.00)

$$\because \int_a^x e^{f(t)} dt = \int_0^x g(x-1) dt + 2x + 3$$

$$\Rightarrow \int_a^x e^{f(t)} dt = \int_0^x g(t) dt + 2x + 3$$

(Using King Property)

Differential both sides, we get

$$e^{f(x)} = g(x) + 2$$

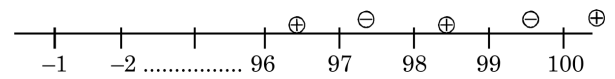
$$\Rightarrow g(x) = e^{f(x)} - 2$$

$$\Rightarrow g'(x) = e^{f(x)} \cdot f'(x)$$

$\because e^{f(x)}$ is always greater than zero.

$\therefore$ Sign of $g'(x)$ is same as sign of $f'(x)$.

$\therefore$ Sign of $g'(x)$



Clearly, local extremum (maximum or minimum) will occur at

$$x = 99, 97, 95, \dots, 3, 1$$

$\therefore$ Sum of all the values = $1 + 3 + 5 + \dots + 99$

$$= \frac{50}{2} [2 \times 1 + (50 - 1) \times 2] = 2500$$

4. Ans (6.00)

From the given equation we have,

$$2 \sin \frac{x+y}{2} \cos \frac{x-y}{2} = 2 \sin \frac{x+y}{2} \cos \frac{x+y}{2}$$

$$\Rightarrow \sin \frac{x+y}{2} \left[\cos \frac{x-y}{2} - \cos \frac{x+y}{2} \right] = 0$$

$$\Rightarrow \sin \frac{x+y}{2} \times 2 \sin \frac{x}{2} \sin \frac{y}{2} = 0$$

Which holds if either $\sin \frac{x+y}{2} = 0$ or $\sin \frac{x}{2} = 0$ or $\sin \frac{y}{2} = 0$

Also since $|x| + |y| = 1 \Rightarrow |x| \leq 1, |y| \leq 1$. So, the required solution is either $x + y = 0$ or $x = 0$ or $y = 0$

If $x = 0, y = \pm 1$, and if $y = 0, x = \pm 1$ and if $x + y = 0$

$$\Rightarrow y = -x \Rightarrow |y| = |-x| = |x| = \frac{1}{2}$$

So that the required pairs (x, y) are

$$(0, \pm 1), (\pm 1, 0), \left(\frac{1}{2}, -\frac{1}{2} \right), \left(-\frac{1}{2}, \frac{1}{2} \right) \text{ which are 6}$$

in number.

5. Ans (2.00)

$$9A^2 - 6A = \begin{bmatrix} 6 & 27 \\ -10 & -1 \end{bmatrix} B^{-1}$$

$$\Rightarrow (3A - I)^2 = \begin{bmatrix} -6 & 27 \\ -3 & 0 \end{bmatrix}$$

$$\Rightarrow (3A - I)^2 = 81$$

$$\Rightarrow |3A - I| = \pm 9 \quad (1)$$

$$\text{Let } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\therefore 9(ad - bc) = 18 \text{ or } 0 \text{ (using equation 1)}$$

Since $A \neq 0$

$$\therefore ad - bc = 2 \Rightarrow |A| = 2$$

6. Ans (4.00)

$$x^2 + (2 - \tan \theta)x - (1 + \tan \theta) = 0$$

$$\alpha + \beta = \tan \theta - 2$$

$$\alpha\beta = -\tan \theta - 1$$

From equation (1) + (2)

$$\alpha + \beta + \alpha\beta = -3$$

$$(\alpha + 1)(\beta + 1) = -2$$

Hence either $\alpha + 1 = -2$ and $\beta + 1 = 1$

then $\alpha = -3, \beta = 0$

or $\alpha + 1 = -1$ and $\beta + 1 = 2$

$\alpha = -2$ and $\beta = 1$

If $\alpha = -3, \beta = 0$ then $\tan \theta = -1$

$$\Rightarrow \alpha = \frac{3\pi}{4}, \frac{7\pi}{4}$$

If $\alpha = -2, \beta = 1$ then $\tan \theta = 1$

$$\Rightarrow \theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$\text{Sum} = \frac{3\pi}{4} + \frac{7\pi}{4} + \frac{\pi}{4} + \frac{5\pi}{4} = 4\pi = 4$$